Phase 8 — BeatAhead Ischemic Stress Index (ISI) Specification & Audit Report

Document ID: BEATAHEAD-PHASE8-ISI-SPECIFICATION
Execution Timestamp: 2026-09-22 19:48:00 UTC
Environment: BeatAhead Specification & Architectural Audit
Phase Status: SPECIFICATION ONLY (Strictly Non-Executable & Non-Modifying)
Upstream Frozen Artifacts: - Frozen Model: `models/beatahead_phase5_model.joblib` (SHA256 verified) - Frozen Schema: `models/feature_schema.json` (Schema v1.0.0, 26 Matrix A features) - Frozen Model Metadata: `models/model_metadata.json` - Frozen Calibrated Threshold: $\tau = 0.156742$ - Frozen Dataset: `data/processed/vitaldb_100_windows.pkl.gz` (100-case VitalDB surgical benchmark)

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1. Executive Summary & Regulatory Boundary

Phase 8 defines the formal engineering, mathematical, and conceptual specification governing how BeatAhead's user-facing **0–100 Ischemic Stress Index (ISI)** relates to the upstream **Frozen Phase 5/6 Machine Learning Inference Engine**.

1.1 Invariants Upheld in Phase 8

In strict compliance with project constraints: 1. **Zero Retraining or Alteration**: The frozen XGBoost model artifact, hyperparameters, 26 Matrix A features, and calibrated decision threshold (0.156742) were untouched. 2. **Zero Code Modification**: No application code, UI components, API routes, or simulation routines (`calculateISI()`) were altered. 3. **Formal Decoupling ($p_{\text{model}} \neq \text{ISI}$)**: The specification explicitly establishes that the XGBoost conditional model probability (p_{model}) is **not** the ISI product score. Direct linear scaling ($\text{ISI} = p_{\text{model}} \times 100$) is formally rejected because it collapses 99% of normal physiology into the $0.0–1.5$ range and distorts user perception. 4. **Strict Non-Diagnostic Positioning**: The ISI is formally defined as an exploratory, illustrative product-level index for trend tracking and research demonstration. It does not constitute a clinically validated risk score, does not predict myocardial infarction, and carries strict prohibitions against diagnostic terminology.

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2. Audit of Existing BeatAhead ISI Implementation

An exhaustive audit of the existing codebase (`C:\ischemic\src\lib\isi\` and related UI components) was conducted to baseline current behavior.

2.1 Current `calculateISI()` Routine (`src/lib/isi/scoring.ts`)

```
export function calculateISI(input: ScoringInput): ISIScore {
  const { features, baseline, historicalScores, scenario, signalQuality, timestamp } = input;

  const hrvFactor = normalizeDeviation(features.hrv.deviation, -15, 15);
  const pulseFactor = normalizeDeviation(features.pulseMorphology.deviation, -10, 20);
  const spo2Factor = normalizeDeviation(features.spo2Trend.deviation, -3, 3);
  const ecgFactor = normalizeDeviation(features.ecg.deviation, -10, 25);
  const motionFactor = features.imu.artifactDetected ? 0.8 : Math.min(features.imu.motionIntensity, 0.5);

  const contributions: ISIContributions = {
    hrv: hrvFactor * 0.28,
    pulseMorphology: pulseFactor * 0.22,
    spo2Trend: spo2Factor * 0.18,
    ecg: ecgFactor * 0.2,
    motionArtifact: motionFactor * 0.12,
  };

  const rawScore =
    baseline.isi +
    contributions.hrv * 35 +
    contributions.pulseMorphology * 30 +
    contributions.spo2Trend * 25 +
    contributions.ecg * 35 +
    contributions.motionArtifact * 20;

  const scenarioModifier = getScenarioModifier(scenario, historicalScores.length);
  const adjustedScore = rawScore * scenarioModifier;
  const score = Math.round(Math.max(0, Math.min(100, adjustedScore)));
  ...
}
```

2.2 Audit Findings & Observations

1. **Inputs**: - `features` : Mocked `FeatureSet` containing HRV SDNN, PPG pulse morphology, SpO2 trend, ECG heart rate, and IMU motion intensity. - `baseline` : `PersonalBaseline` with default values: resting HR: 68 bpm, HRV: 51 ms, SpO2: 97%, pulse morphology: 0.85, baseline ISI: 48. - `historicalScores` : Array of recent numeric scores. - `scenario` : Simulated test state (`"normal" | "stress_event" | "recovering" | "persistent_rising" | "motion_artifact"`). - `signalQuality` : Single scalar percentage ($0–100$).
2. **Normalization & Weighting**: - Feature deviations are clamped and linearly normalized to $[0, 1]$ via `normalizeDeviation(val, min, max)` . - Weights: HRV (28%), Pulse Morphology (22%), SpO2 Trend (18%), ECG (20%), Motion Artifact (12%). Total $= 100\%$. - Scale multiplier: Weighted contributions are scaled by factors between 20 and 35 and added to `baseline.isi` (48).
3. **Simulation Modifiers**: - Scores are modulated by synthetic sinusoidal multipliers (`getScenarioModifier`) based on tick count.
4. **Current Thresholds & Labels (`src/lib/utils.ts`)**: - `0 - 30` : `"Lower observed trend"` (Green `#10B981`) - `31 - 60` : `"Intermediate observed trend"` (Amber `#F59E0B`) - `61 - 100` : `"Higher observed trend"` (Red `#DC2626`) - Risk Trend Banner: Triggers when $\text{score} > \text{baseline.isi} + 10$ (> 58 points).
5. **Existing UI References**: - **`ISIGauge.tsx`** : Radial circular gauge displaying 0–100 integer score, personal baseline (48), current deviation, trend arrow, confidence percentage, and disclaimer tooltip. - **`RiskTrendBanner.tsx`** : Context banner displaying whether physiology is within or above recent personal baseline. - **`ContributionBars.tsx`** : Horizontal progress bars showing relative influence of HRV, SpO2, pulse

morphology, and motion. - `Live Monitor` (`/monitor`): Integrates real-time gauge with rolling line chart. - `Trends` (`/trends`): 24h, 7-day, and 30-day historical time-series graphs.

6. **Codebase Disconnect**: - The comment in `scoring.ts` states: "*Replace this function with a real XGBoost model inference call in production.*" - This audit confirms that simply swapping `calculateISI()` with raw model probability would break the UI: an intraoperative ischemia probability of 0.005 (normal surgical state) would display an ISI of 0 or 1, rendering the 48-baseline gauge completely unresponsive.

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3. Five Conceptual Layer Definitions

To establish modularity and clinical clarity, BeatAhead defines five strictly separated layers:

Layer A: Frozen Model Output

- **Nature**: Statistical binary classifier (XGBoost) trained on 100 surgical cases.
- **Inputs**: 26 Matrix A multimodal features computed over $T_{\text{obs}} = 300\text{ s}$.
- **Outputs**:
- $p_{\text{model}} \in [0, 1]$: Statistical likelihood of prospective intraoperative ST depression event (≥ 60s at $\text{ST} \le -1.0\text{ mm}$ after a 300s buffer).
- $\tau_{\text{frozen}} = 0.156742$: Frozen calibrated decision threshold.
- $\text{ALERT} \in {0, 1}$: Binary state ($\text{ALERT} = 1 \iff p_{\text{model}} \ge 0.156742$).

Layer B: Signal Quality Assessment (SQA)

- **Nature**: Multi-sensor reliability and integrity gating engine.
- **Components**:
- $Q_{\text{ecg}} \in [0, 1]$: Clean QRS detection fraction.
- $Q_{\text{ppg}} \in [0, 1]$: Waveform clipping and perfusion signal quality index.
- $V_{\text{pat}} \in {0, 1}$: Biologically plausible pulse arrival transit flag (100–450 ms in $\ge 30\%$ beats).
- $I_{\text{motion}} \in [0, 1]$: Inertial accelerometer disturbance magnitude.
- **Role**: Quality gating prevents noisy sensor data from masquerading as physiological deterioration.

Layer C: Personal Baseline Tracking

- **Nature**: Patient-specific resting coordinate system.
- **Parameters**: Baseline heart rate (HR_{base}), variability ($SDNN_{\text{base}}$), transit (PAT_{base}), oxygenation ($SpO2_{\text{base}}$), and ST level (ST_{base}).
- **Role**: Quantifies relative physiological disturbance (Δ). Disclaimed as non-predictive.

Layer D: Temporal Trend & Momentum

- **Nature**: Rate of change and trajectory velocity tracker.
- **Metrics**: 5-minute regression slope (v), 30-minute cumulative trajectory, and discrete trend direction (`increasing`, `decreasing`, `stable`).
- **Role**: Differentiates acute transient fluctuations from sustained physiological drift.

Layer E: ISI Product Score

- **Nature**: 0–100 composite product index.

- **Synthesis**: Combines Model Evidence (E_{model}), Autonomic Deviation (D_{auto}), Perfusion Strain (D_{perf}), and Trend Momentum (M_{trend}) anchored to personal resting baseline.
- **Role**: Provides a responsive, easily interpreted wellness and stress indicator for user interface surfaces.

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4. Formal ISI Input Specification

| Input | Source | Units | Normalization | Direction | Missing Data Behavior | Quality Dependency | | :--- | :--- | :--- | :--- | :--- | :--- | :--- | :--- | | **p_{model}** | XGBoost Matrix A | Probability $[0, 1]$ | Piecewise Sigmoidal | Higher $\rightarrow$ Higher ISI | Hold last valid up to 120s; fallback to heuristic | Gated if $Q_{\text{ecg}} < 0.50$ | | **ALERT** | Threshold Comparator | Binary ${0, 1}$ | ${0, 1}$ at $\tau=0.156742$ | $1 \rightarrow$ Elevates ISI Floor | Return 0 if model offline | Inherits p_{model} dependency | | **ΔHR_{base}** | ECG Lead II | bpm | $\frac{HR - HR_{\text{base}}}{HR_{\text{base}}} \times 100$ | Higher $\rightarrow$ Higher ISI | Assume $\Delta = 0$ | Clamped if $Q_{\text{ecg}} < 0.60$ | | **$\Delta SDNN_{\text{base}}$** | ECG Lead II | ms | $\frac{SDNN - SDNN_{\text{base}}}{SDNN_{\text{base}}} \times 100$ | Lower $\rightarrow$ Higher ISI | Assume $\Delta = 0$ | Suppressed if ectopic $> 15\%$ | | **ΔPAT_{base}** | ECG + PPG Synchrony | ms | $\frac{PAT - PAT_{\text{base}}}{PAT_{\text{base}}} \times 100$ | Shorter $\rightarrow$ Higher ISI | Exclude from factor pool | Omitted if $V_{\text{pat}} == 0$ | | **$\Delta SpO2_{\text{base}}$** | Pulse Oximeter | % | $\min(0, SpO2 - SpO2_{\text{base}})$ | Lower $\rightarrow$ Higher ISI | Assume $\Delta = 0$ | Clamped if $Q_{\text{ppg}} < 0.60$ | | **ΔST_{obs}** | ECG Lead II | mm | $ST_{\text{median}} - ST_{\text{base}}$ | Depressed $\rightarrow$ Higher ISI | Fallback to non-ST heuristic | Gated if Lead II detached | | **Q_{ecg}** | QRS Detector | Ratio $[0, 1]$ | Linear $[0, 1]$ | Higher $\rightarrow$ Higher Confidence | Defaults to 0.0 (Unusable) | Direct quality sensor | | **Q_{ppg}** | PPG Pulse Analyzer | Ratio $[0, 1]$ | Linear $[0, 1]$ | Higher $\rightarrow$ Higher Confidence | Defaults to 0.0 (Unusable) | Direct quality sensor | | **V_{pat}** | Pulse Transit Engine | Binary ${0, 1}$ | ${0, 1}$ | $1 \rightarrow$ Valid transit | Defaults to 0 (Invalid) | Synchrony dependency | | **I_{motion}** | Accelerometer | a.u. / g | Clamped $[0, 1]$ | Higher $\rightarrow$ Lower Confidence | Defaults to 0.0 | Inertial sensor | | **T_{velocity}** | Moving Regression | points/min | Bounded $[-5, +5]$ | Positive $\rightarrow$ Rising Trend | 0.0 during initial 5 min | Requires ≥ 5 clean ticks |

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5. Personal Baseline Architecture

1. **Initialization**: - Minimum **600 seconds (10 minutes)** of continuous monitoring. - Minimum data validity: $\ge 80\%$ of windows with $Q_{\text{overall}} \ge 0.70$. - System state during acquisition: `Baseline Establishing` (displays acquisition percentage; no premature scores).
2. **Adaptive Rolling Update (EMA)**: $$B_t = (1 - \alpha) B_{t-1} + \alpha X_t \quad (\alpha = 0.02)$$
3. **Anti-Drift Gating Rule (Safety Critical)**: $$\text{Update Allowed} \iff (\text{State} == \text{"Normal / Stable"}) \land (Q_{\text{overall}} \ge 0.75) \land (I_{\text{motion}} < 0.30)$$ **Baseline updates are strictly locked during `Elevated Model Evidence` or `Elevated ISI Trend`.** This prevents chronic physiological stress or acute ischemia from shifting the baseline upward and masking pathology.
4. **Outlier Rejection**: Values outside $\pm 3.0 \times \text{IQR}$ are rejected as transient motion or sensor detachment.

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6. Mathematical Prototype Specification

6.1 Model Evidence Mapping (E_{model})

To bridge the frozen decision threshold ($\tau = 0.156742$) to the 0–100 scale:

$$E_{\text{model}}(p) = \begin{cases} 0.50 \times \left(\frac{p}{\tau} \right)^{1.4} & p < \tau \ 0.50 + 0.50 \times \left(1 - \exp\left(-4.0 \frac{p - \tau}{1 - \tau} \right) \right) & p \ge \tau \end{cases}$$

- $p = 0.005$ (normal surgical baseline) $\rightarrow E_{\text{model}} \approx 0.008$
- $p = \tau = 0.156742$ (exact decision threshold) $\rightarrow E_{\text{model}} = 0.500$
- $p = 0.300 \rightarrow E_{\text{model}} \approx 0.77$
- $p = 0.600 \rightarrow E_{\text{model}} \approx 0.96$

6.2 Autonomic Shift Factor (D_{auto})

$$f_{\text{hr}} = \text{clamp}\left(\frac{HR - HR_{\text{base}}}{25.0}, 0.0, 1.0 \right)$$ $$f_{\text{hrv}} = \text{clamp}\left(\frac{SDNN_{\text{base}} - SDNN}{0.60 \times SDNN_{\text{base}}}, 0.0, 1.0 \right)$$ $$D_{\text{auto}} = 0.50 \cdot f_{\text{hr}} + 0.50 \cdot f_{\text{hrv}}$$

6.3 Oxygenation & Perfusion Factor (D_{perf})

$$f_{\text{spo2}} = \text{clamp}\left(\frac{SpO2_{\text{base}} - SpO2}{5.0}, 0.0, 1.0 \right)$$ $$f_{\text{pat}} = \text{clamp}\left(\frac{PAT_{\text{base}} - PAT}{50.0}, 0.0, 1.0 \right) \quad (\text{if } V_{\text{pat}} = 1, \text{ else omitted})$$ $$D_{\text{perf}} = 0.60 \cdot f_{\text{spo2}} + 0.40 \cdot f_{\text{pat}} \quad (\text{or } D_{\text{perf}} = f_{\text{spo2}} \text{ if PAT invalid})$$

6.4 Trend Momentum (M_{trend})

$$M_{\text{trend}} = \text{clamp}\left(\frac{v}{2.0}, -0.15, +0.15 \right) \quad (v = \text{points / minute over 5 min})$$

6.5 Composite Synthesis & Guaranteed Bounding

$$\text{ISI}_{\text{raw}} = 40.0 + \left[0.45(E_{\text{model}} - 0.20) + 0.30 D_{\text{auto}} + 0.25 D_{\text{perf}} + M_{\text{trend}} \right] \times 60.0$$ $$\text{ISI} = \text{round}\left(\max\left(0.0, \min\left(100.0, \text{ISI}_{\text{raw}} \right) \right) \right)$$

$$\forall \text{ valid physiological vectors}, \quad 0 \le \text{ISI} \le 100$$

7. Model Alert State vs. ISI Display Score

Attribute	Model Alert State (ALERT)	ISI Display Score (ISI)
Output Type	Discrete Binary Flag (${0, 1}$)	Bounded Continuous Index ($[0, 100]$)
Threshold	Frozen Calibrated Value: **0.156742**	Low (≤ 30), Mid ($31–60$), High (≥ 61)
Meaning	Machine learning forecast of prospective intraoperative ST depression event ≥ 60s	Multi-factor physiological strain index relative to baseline
UI Placement	Research Model Signal Card (`/insights`)	Circular Gauge (`/monitor` , `/dashboard`)

8. Signal Quality & Degraded Sensor Behavior

$$Q_{\text{overall}} = Q_{\text{ecg}} \times 0.40 + Q_{\text{ppg}} \times 0.35 + V_{\text{pat}} \times 0.15 + (1 - I_{\text{motion}}) \times 0.10$$

1. **$Q_{\text{ecg}} < 0.50$**: Suppress model inference; freeze ST & HRV terms; display `"Signal Degraded"` .
2. **$Q_{\text{ppg}} < 0.50$**: Suppress PPG morphology and PAT terms; calculate score solely from ECG and SpO2.
3. **$V_{\text{pat}} == 0$**: Exclude PAT from perfusion term gracefully without raising errors.
4. **Missing SpO2**: Retain nominal $SpO2_{\text{base}}$, reduce overall confidence by 20%.
5. **$I_{\text{motion}} > 0.70$**: Lock score buffer; display `"Motion Artifact Detected"` .
6. **$Q_{\text{overall}} < 0.35$**: System enters `Insufficient Signal Quality` ; gauge displays `--` .

9. Five Product Output States

1. `Insufficient Signal Quality` : $Q_{\text{overall}} < 0.35$. Gauge shows `--` .
2. `Baseline Establishing` : Monitoring time $< 600\text{ s}$. Progress percentage shown.
3. `Normal / Stable` : $Q_{\text{overall}} \ge 0.50$, $p_{\text{model}} < 0.156742$, $\text{ISI} \le 60$. Green or amber gauge. Label: `"Lower/Intermediate observed trend"` .
4. `Elevated Model Evidence` : $p_{\text{model}} \ge 0.156742$. Explicit alert callout: `"Early-warning model evidence elevated"` .
5. `Elevated ISI Trend` : $\text{ISI} \ge 61$ due to cumulative autonomic/perfusion drift even if model is sub-threshold. Red gauge.

> [!CAUTION] **Prohibited Diagnostic Terminology:**
> The system must NEVER state: "*Heart attack detected*", "*Ischemia confirmed*", "*Myocardial infarction occurring*", or "*Coronary artery occluded*". All language must remain strictly observational and trend-oriented.

10. AI Insights Explainability Boundaries

- **Permitted**:

- Explaining whether the machine learning model detected physiological patterns matching surgical benchmark early-warning signatures.
- Reporting percentage deviations from the individual's resting baseline.
- Presenting relative feature contribution breakdowns (HRV, PAT, SpO2, ST).
- Highlighting sensor quality or motion artifacts reducing confidence.
- **Strictly Forbidden**:
- Formulating clinical diagnoses or ruling out pathology.
- Recommending medication adjustments or cessation.
- Reassuring users to ignore acute chest pain, shortness of breath, or radiating arm pain.

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11. Delivery Verification Checklist

- [x] Full audit of existing BeatAhead ISI codebase completed and documented.
- [x] Conceptual separation of 5 layers established.
- [x] Formal ISI input specification table generated (12 tracked inputs).
- [x] Personal baseline architecture specified (initialization, anti-drift gating, outlier rejection).
- [x] Mathematical prototype specification established with guaranteed $[0, 100]$ bounds.
- [x] Model decision threshold (0.156742) strictly preserved.
- [x] Signal quality degradation rules defined.
- [x] 5-state product automaton defined with non-diagnostic terminology rules.
- [x] AI Insights explainability boundaries defined.
- [x] Deliverables created: `docs/ISI_SPECIFICATION.md`, `docs/ISI_DATA_FLOW.md`, `reports/phase8_isi_specification.md`.
- [x] Zero code, UI, API, model artifact, or dataset modifications made.